Analysis of Symmetry of 2-point Correlation Functions in Finite Temperature QCD

Using Möbius Domain-Wall Fermions

Junxiong Nie, U. Osaka
for JLQCD Collaboration

Outline

- Research Background
- Theoretical Background
- Lattice Simulation Setup
- Numerical Results
- Summary and Outlook

Research Background

- Chiral phase transition, which occurred in the early universe, is key to particle physics,nuclear physics, and cosmology
- The behavior of related symmetries determines the order and universality class
- Previous lattice studies used algorithms that break chiral symmetry

Motivation for this work

- We use Möbius Domain-Wall Fermions (MDWF) to respect chiral symmetry
- Our recent study by JLQCD collaboration investigated $N_f = 2$ MDWF [PRD 111, 114506 (2025)]
- Now we extend the simulation by adding the strange quark ($N_f = 2 + 1$)

This work

- We extend this investigation to a more realistic $N_f = 2 + 1$ QCD case adding strange quark and improve it to finer lattice size up to $\mathrm{L}48^3 \times \mathrm{T}16$ ($L_s = 12$)
- We study the fixed $\beta = 4.17$ case with 4 values of quark masses to investigate the quark mass dependence, and the physical point case varying beta.
- We measure spatial two point correlation function data and analysed the symmetry restoration behavior for physical point with different beta and chiral limit case by a fixed beta case with varying quark masses and lattice size

  [Phys. Rev. D 111, 114506 (2025)]

Symmetry of QCD

QCD: $\mathcal{Z} = \int \mathcal{D}A\mathcal{D}\bar{q}\mathcal{D}q e^{-S}$, Gauge Boson+Fermion

$S = \int d^4x \left[-\frac{1}{2g^2} \mathrm{Tr} F^{\mu\nu} F_{\mu\nu} + \bar{q}_a (i\not{D} - m_a) q_a \right]$, a in [1,N_f]

Specifically,in the up-down quark mass zero limit,
Flavored chiral symmetry $SU(2)_L \times SU(2)_R$:

$$q \to \exp(i\alpha_a T^a + i\beta_a T^a \gamma_5) q, \quad \bar{q} \to \bar{q} \exp(-i\alpha_a T^a + i\beta_a T^a \gamma_5),$$

$U(1)_A$ Symmetry :

$$q \to \exp(i\epsilon\gamma_5) q, \quad \bar{q} \to \bar{q} \exp(i\epsilon\gamma_5),$$

Possible Emergent Larger Symmetry

- Perturbative analysis of screening masses [Laine & Vepsäläinen (2004)], predicts for the $N_f = 2$ case a **spin-independent** result of the meson screening masses at high T due to a large Matsubara mass::

$$m_{\rm scr} = 2\pi T + 0.1872 g^2 T$$

- $\mathrm{U}(1)_A$ will be restored at high temperature; also predicted by semi-classical analysis of anomaly effect
- Inspire a larger symmetry, Chiral-Spin Symmetry $SU(2)_{CS}$, may emerge:

$$q \to \exp(i\Sigma^i \epsilon_i) q,$$
$$\bar{q} \to \bar{q}\gamma_0 \exp(-i\Sigma^i \epsilon_i)\gamma_0, \qquad \text{where } \Sigma^i = (\gamma_k, -i\gamma_5\gamma_k, \gamma_5), \quad k = 1, 2.$$

Meson Correlation Functions

We consider the following set of meson operators:

$$O_\Gamma(x) = \bar{q}(x)\Gamma F^a q(x),$$

$$\Gamma \in \left\{ \mathbf{1}(S), \gamma_5(P), \gamma_\mu(V), \gamma_\mu\gamma_5(A), \sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu](T), \sigma_{\mu\nu}\gamma_5(X) \right\}$$

And we measure the two-points correlation function of the meson operators of different channels. Accroding to the symmetry properties, the asymptotic behavior of the 2-point correlation function at large distances should be approximately be $A\mathrm{Cosh}(\mathrm{m}_\Gamma(\mathrm{x} - \mathrm{L}/2))$

Symmetry of Correlation Functions

- Flavored chiral symmetry $SU(2)_L \times SU(2)_R$: $\delta_C O^a_\Gamma = \frac{i}{2}\beta_b \bar{q}\left(\{\Gamma,\gamma_5\}\frac{1}{2}\delta^{ab}\mathbf{1} + [\Gamma,\gamma_5] i\epsilon^{bac}\frac{\tau^c}{2}\right) q$
- $U(1)_A$ Symmetry : $\delta_A O^a_\Gamma = i\epsilon\bar{q}\{\Gamma,\gamma_5\}\frac{\tau^a}{2} q$

Γ	$[\Gamma,\gamma_5]$	$\{\Gamma,\gamma_5\}$
$\mathbf{1}(S)$		γ_5
$\gamma_5(P)$		$\mathbf{1}$
$\gamma_\mu(V)$	$\gamma_\mu\gamma_5$	
$\gamma_\mu\gamma_5(A)$	γ_μ	
$\sigma_{\mu\nu}(T)$		$\sigma_{\mu\nu}\gamma_5$
$\sigma_{\mu\nu}\gamma_5(X)$		$\sigma_{\mu\nu}$

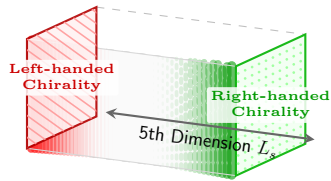
Γ	Reference Name	Abbr.	Symmetry Correspondences		
$\mathbb{I}$	Scalar	S	$U(1)_A$		
γ_5	Pseudo Scalar	PS			
γ_1,γ_2	Vector	V	$SU(2)_L \times SU(2)_R$		$SU(2)_{CS}$
$\gamma_1\gamma_5,\gamma_2\gamma_5$	Axial Vector	A			
$\gamma_4\gamma_3$	Tensor	T_t	$U(1)_A$		
$\gamma_4\gamma_3\gamma_5$	Axial Tensor	X_t			

Domain-Wall Fermions (DWF)

- **Localized Chiral States**: DWF realizes chiral fermions as boundary-localized states
- **Residual Mass and Chiral Breaking**: At finite L_s, chiral symmetry is broken by:

$$\overline{\psi}\Delta_L\psi \simeq m_{\text{res}}\overline{\psi}\psi + O(A,\bar{\psi},\psi) + \mathcal{O}(a^2)$$

$$\delta_{\text{chiral}} S_{4\text{D}} = i\alpha m_q \bar{\psi}(\gamma_5 + \hat{\gamma}_5)\psi + 2i\alpha\bar{\psi}\gamma_5\Delta_L\psi$$
$$\Delta_L = \frac{1}{4}(1-\epsilon_L^2)$$



$$D^{5\text{D}}_{s,s'}(m) = D^{(s)}_{+}\delta_{s,s'} + D^{(s)}_{-}(P_+\delta_{s-1,s'} + P_-\delta_{s+1,s'})$$
$$- m(D^{(1)}_{-}P_+\delta_{s,1}\delta_{s',L_s} + D^{(L_s)}_{-}P_-\delta_{s,L_s}\delta_{s',1})$$

where $D^{(s)}_{+} = b_5 D_W + 1$, $D^{(s)}_{-} = c_5 D_W - 1$,
$P_\pm = \frac{1\pm\gamma_5}{2}$ $(b_5 = 1.5, c_5 = 0.5)$.

Residual Mass Measurement

We measure the residual mass am_{res} using the ratio of two-point correlation functions:

$$am_{\text{res}} = \frac{\sum_{\vec{x}} \langle J_{5q}(\vec{x}, t) P(\vec{0}, 0) \rangle}{\sum_{\vec{x}} \langle P(\vec{x}, t) P(\vec{0}, 0) \rangle}$$

T (MeV)	138	145	153	157	164.5	176	202	241.6
m_{res}/m_l	0.9087	0.5360	0.3394	0.2744	0.1836	0.1041	0.0299	0.0049
m_{res} (MeV)	1.61	1.16	0.832	0.703	0.503	0.304	0.0908	0.0145

Lattice Setup & Configurations

- Physical Point: Lattice size $48^3 \times 16$, with 16-source measurements averaging for all channels and take the averaging for all symmetric spatial directions.
- Fixed $\beta = 4.17$: Varying light quark mass am_l and temporal size N_t to study the chiral limit, with 16 source measurement and spatial averaging too.

Physical Point Setup ($N_f = 2 + 1$)

β	T (MeV)	am_s	am_l	N_{conf}	Total Meas.
4.13	138	0.043547	0.000805	395	6320
4.15	145	0.040843	0.000930	398	6368
4.17	153	0.038400	0.001001	396	6336
4.18	157	0.037265	0.001022	402	6432
4.20	164.5	0.035150	0.001041	407	6512
4.23	176	0.032315	0.001033	395	6320
4.30	202	0.026930	0.000939	404	6464

Fixed $\beta = 4.17$ Setup

Lattice Size	am_s	am_l	T (MeV)
$32^3 \times 12$	0.040	0.0020, 0.0035, 0.0070, 0.0120	220.00
$32^3 \times 14$	0.040	0.0020, 0.0035, 0.0070, 0.0120	188.57
$32^3 \times 16$	0.040	0.0020, 0.0035, 0.0070, 0.0120	165.00
$36^3 \times 18$	0.040	0.0020, 0.0035, 0.0070, 0.0120	146.67

Extracting Screening Masses: Methodology

- Screening Mass Fit: The physical screening mass m_{scr} is extracted by performing a least-squares χ^2 fit of $C(x)$ to the asymptotic form:

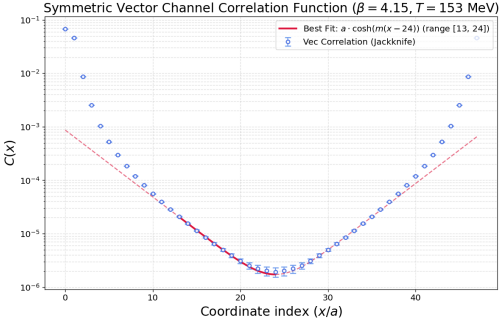
$$C(x) = A \cosh\left(m_{\text{scr}}\left(x - \frac{L}{2}\right)\right)$$

- Local Effective Mass: The local effective mass $m_{\text{eff}}(x)$ is solved numerically to identify a suitable plateau, which is used to determine the fitting range $[x_{\text{min}}, x_{\text{max}}]$:

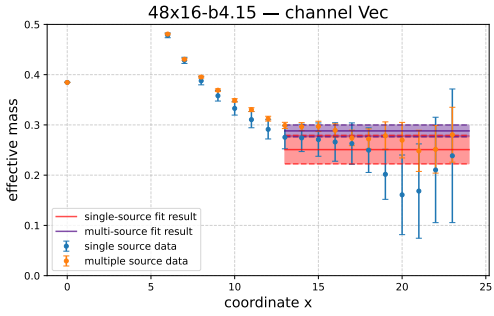
$$\frac{C(x)}{C(x+1)} = \frac{\cosh(m_{\text{eff}}(x)(x - L/2))}{\cosh(m_{\text{eff}}(x)(x + 1 - L/2))}$$

- Statistical Error: Jackknife method and Binning analysis are used to handle statistical uncertainties and eliminate HMC autocorrelation bias.

Extracting Screening Masses: Fit Examples

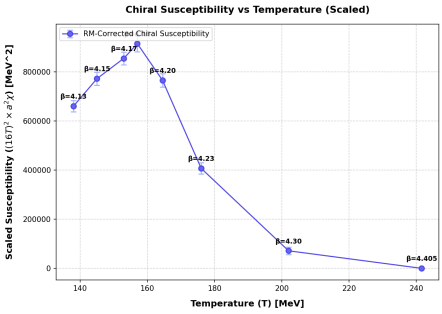


correlation data of Vec channel for $\beta = 4.15$(T=153MeV)



effective mass of Vec channel for $\beta = 4.15$(T=153MeV)

Disconnected Chiral Susceptibility



Disconnected susceptibility χ_{dis} vs Temperature

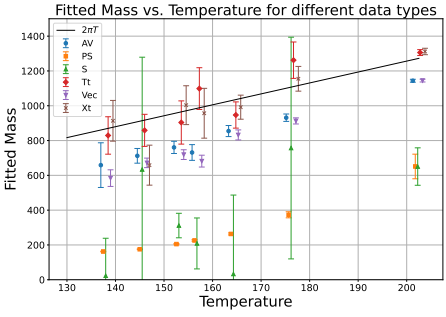
- Definition:

$$a^2\chi_{\text{dis}} = \frac{1}{N_s^3 N_\tau}\left(\frac{N_f}{4}\right)^2 \cdot \left(\langle(\mathrm{Tr}\not{D}^{-1})^2\rangle - \langle\mathrm{Tr}\not{D}^{-1}\rangle^2\right)$$

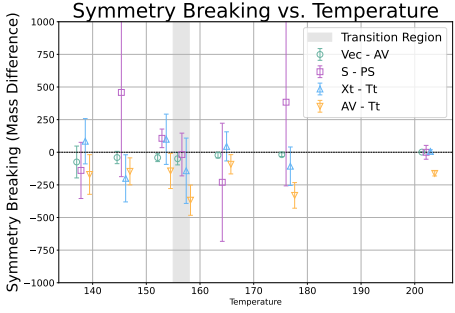
- Fluctuations of order parameter, related to topological effects.
- Clear peak determines the pseudo-critical temperature, consistent with the HotQCD result(2019) :$T_c = 156.5 \pm 1.5$ MeV.

Numerical Results

Grey band is 156 MeV $\pm$ 1.5 MeV



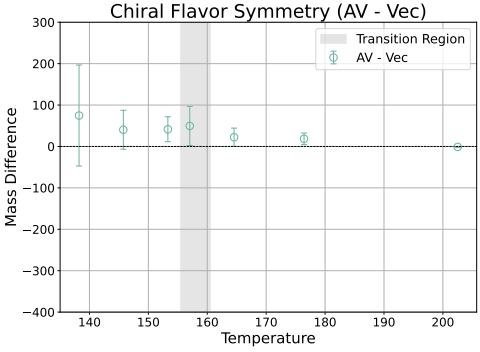
simulated effective masses of different channels



Symmetry breaking of different channels

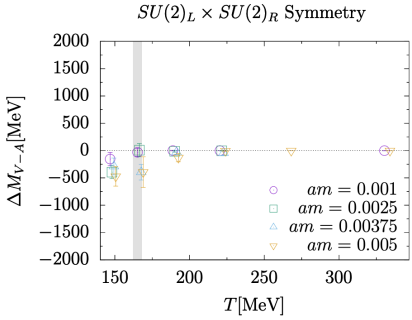
$SU(2)_L \times SU(2)_R$ Chiral Symmetry

Our Result ($N_f = 2 + 1$ MDWF)



Restoration behavior of chiral symmetry

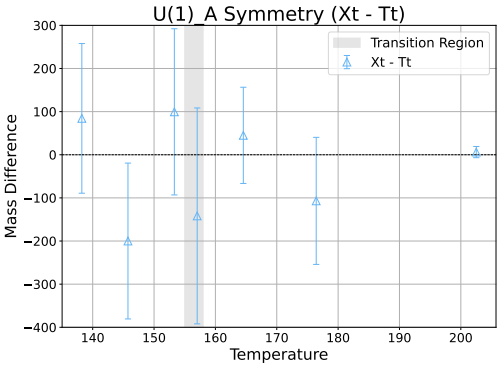
Previous Result ($N_f = 2$ MDWF)



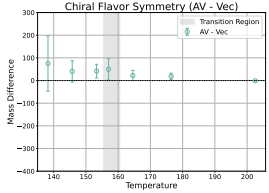
For comparison [PRD 111, 114506 (2025)]

$U(1)_A$ Symmetry $(X_t - T_t)$

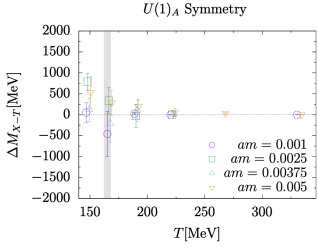
Our $U(1)_A$ ($N_f = 2 + 1$ MDWF)



Our Chiral ($N_f = 2 + 1$ MDWF)

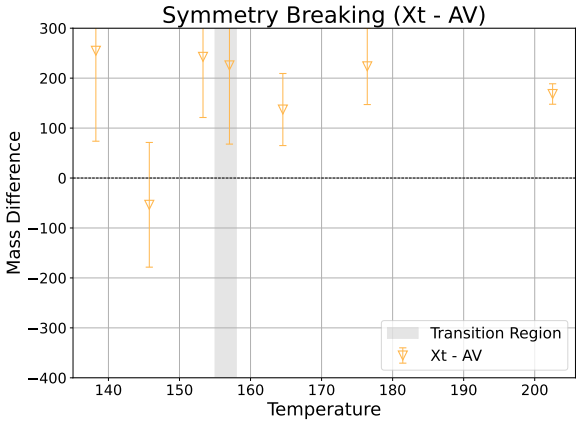


Previous $U(1)_A$ ($N_f = 2$ MDWF)



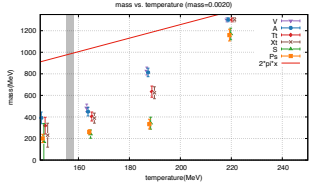
$SU(2)_{CS}$ Symmetry

Broken up to 202 MeV

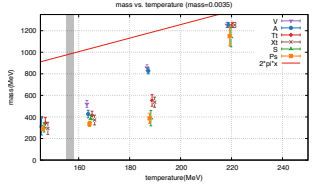


Screening Masses at Fixed $\beta = 4.17$

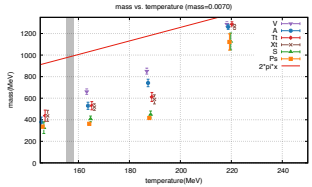
Temperature dependence of screening masses for different light quark masses:



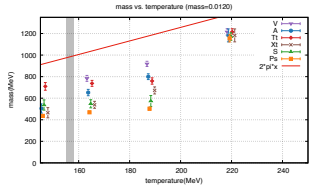
$am_l = 0.0020$



$am_l = 0.0035$



$am_l = 0.0070$

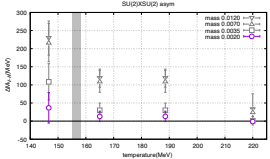


$am_l = 0.0120$

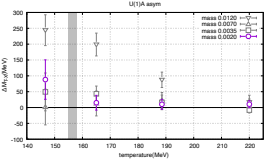
- As temperature increases, partner channels in PS-S and V-AV sectors become degenerate.
- This degeneration is observed for all simulated light quark masses.

Symmetry Restoration at Fixed $\beta = 4.17$

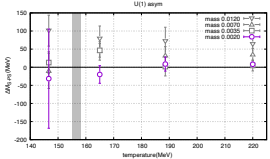
Asymmetry observables as a function of temperature for various light quark masses:



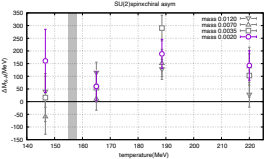
$SU(2)_L \times SU(2)_R$ (V-AV)



$U(1)_A$ (T-X)



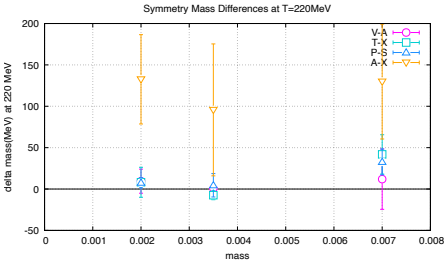
$U(1)$ (S-PS)



$SU(2)_{CS}$ (X-AV)

- All symmetries show a clear trend towards restoration above T_c.
- $U(1)_A$ and chiral $SU(2)_L \times SU(2)_R$ symmetries show an almost simultaneous restoration as we approach the chiral limit.
- Chiral-spin symmetry $SU(2)_{CS}$ remains broken in the analyzed temperature range.

Chiral Limit Extrapolation ($T \approx 220$ MeV)



Mass differences between mesonic partners as a function of am_l at $T \approx 220$ MeV

- Analyzed mass differences at a high-temperature snapshot ($T \approx 220$ MeV) as a function of the bare light quark mass am_l.
- As $am_l \to 0$, the mass differences between chiral partners ($m_{\mathrm{Vec}} - m_{\mathrm{AV}}$ and $m_{\mathrm{S}} - m_{\mathrm{PS}}$) extrapolate toward zero.

Summary and Outlook

- Physical Point: Crossover transition with $T_c \approx 158$ MeV determined by chiral condensation and disconnected susceptibility. Above T_c, chiral flavor and $U(1)_A$ symmetry breakings are consistent with zero within errors. But the scalar channel is nosier than $N_f = 2$ case.
- Chiral Limit Study: Extrapolations to $am_l \to 0$ both of the breaking of chiral flavor and $U(1)_A$ symmetries consistent with zero within large noisy errors.
- Chiral-Spin Symmetry: Remains broken up to 220 MeV
- Outlook: Investigate the large oscillations in the S (Scalar) channel near T_c (potential physical origin); measure susceptibility and higher-order fluctuations, etc.

Backup Slides

Backup: Overlap Dirac Operator

The massless operator $D_{\rm ov}(0) = \frac{1}{2}(1 + \gamma_5 {\rm sgn}(H_W))$ with Hermitian H_W satisfies the Ginsparg-Wilson relation:

$$\gamma_5 D_{\rm ov} + D_{\rm ov}\gamma_5 = 2 D_{\rm ov}\gamma_5 D_{\rm ov}$$

The massive overlap operator is defined as:

$$D_{\rm ov}(m) = D_{\rm ov}(0) + m(1 - D_{\rm ov}(0)) = \frac{1+m}{2} + \frac{1-m}{2}\gamma_5 {\rm sgn}(H_W)$$

$$\widehat{\gamma}_5 = \gamma_5 \left(1 - 2\not{D}_{\rm ov}(0)\right) = -\gamma_5 {\rm sgn}(H_W) \tag{1}$$

which is hermitian and satisfies $\widehat{\gamma}_5^2 = 1$ by virtue of the Ginsparg-Wilson relation. The lattice chiral transformation is defined as:

$$\delta\psi = i\alpha\widehat{\gamma}_5\psi, \quad \delta\overline{\psi} = i\alpha\overline{\psi}\gamma_5 \tag{2}$$

Backup: Order of Chiral Phase Transition

- Perturbative $4-\epsilon$ expansion RG analysis of effective Φ^4 linear sigma model shows no stable IR fixed point for $N_f > \sqrt{3}$ flavor symmetry [Pisarski-Wilczek (1983)].
- For $N_f \geq 3$, the anomaly effect doesn't change the order of the transition; for $N_f = 2$, it becomes a quadratic term which can smooth out the singularity and make the transition 2nd-order.
- If the anomaly effect still exists, the transition can be O(4) class.
- Higher-loop/FRG computations suggest other possibilities [Vicari et al. 2005, Fejos 2022] and Conformal Bootstrap indicates interesting points [Nakayama-Ohtsuki 2016].

Backup: Transition Scenario

Symmetry restoration behavior of $U(1)_A$ and flavor $SU(2)_L \times SU(2)_R$ determines the universality class:

- Previous Scenario: 1st-order if $U(1)_A$ is restored; 2nd-order $O(4)$ if $U(1)_A$ remains broken.
- New Scenario: $O(4)$ if $U(1)_A$ remains broken; weak 1st-order, or 2nd-order ($O(4) \times Z_2$, $O(4) \times Z_4$, $O(4) \times O(2)$) if $U(1)_A$ is restored.

Backup: Summary of Correlation Functions

